INCH-POUND
MIL-PRF-1/1373D
2 December 2008
SUPERSEDING
MIL-E-1/1373C (NAVY)
30 May 1973

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

1/ TYPE 7AQP7 AND 7AQP33

Inactive for new design
after 5 December 1997

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Magnetic deflection, electrostatic focus

PIN CONNECTIONS AND DIMENSIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	EC1	Ec2	Eb2	Ehk	Rg1	Eb1
Unit:	V	V dc	V dc	V dc	V dc	MegΩ	V dc
Maximum:	6.9	0, -180	770	12,100	± 180	Note 1	1,100
Minimum	5.7	----	----	7,000	----	----	-550
Test Conditions	6.3	Adjust	300	10,000	---	---	Focus

GENERAL:
Qualification – required
1/ See note 6

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TABLE I. Testing and inspection.

Inspection	MIL-STD-1311 Method	Notes	Conditions	Symbol	Limits		Unit
					Minimum	Maximum	
Conformance inspection, Part 1							
Voltage breakdown	5201	---		----	-----	-----	-----
Voltage breakdown (electrostatic types)	5201	---		----	-----	-----	-----
Gas ratio	5206	4		Gr	-----	0.25	-----
Bulb, screen, and faceplate quality	5106	---		-----	-----	-----	-----
Modulation	5223	---	Ib2= 100 µAdc	ΔEc1	-----	32	Vdc
Spot position (magnetic deflection)	5231	---		------	-----	0.375	Inch
Grid cutoff voltage	5241	---		Ec1	-15	-45	Vdc
Grid No. 1 leakage current	5251	---		-----	-----	-----	-----
Conformance inspection, part 2							
Focusing electrode voltage	5246	---	Ib2 = 100 µAdc	Eb1	-50	350	Vdc
Heater current	1301	---		If	270	330	mA
Side terminal and bas alinement	5101	---	Halfway between pins No. 1 and No. 9	----	------	-------	-------
Neck and bulb alinement (magnetic types)	5101	5		-------	-------	-------	-------
Face it	5101	----		-------	-------	-------	-------
Neck strightness	5101	2					
Stray light emission (conventional types)	5216	-----	Eb2 = 12,000 Vdc; Ec2 = 770 Vdc; Eb1=0	-------	-------	-------	-------
Screen (p7)	5221	3		-------	-------	-------	-------
Linewidth “A” (magnetic deflection)	5226	---	Ib2 = 100 µAdc	Width	-------	0.38	mm
Heater-cathode leakage current	5221	---		-------	-------	-------	-------
Grid No. 2 leakage Current	5251	---		-------	-------	-------	-------
Base pin solder depth	1111	---		-------	-------	-------	-------
Secureness of base, cap. Or inset	1101	---		-------	-------	-------	-------
Permanence of marking	1105	---		-------	-------	-------	-------

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TABLE I. Testing and inspection. Continued

Inspection	MIL-STD-1311 Method	Notes	Conditions	Symbol	Limits		Unit
					Minimum	Maximum	
Conformance inspection, Part 3							
Life Test	-----		Group C: Eb2 = 12,100 Vdc; Ec2 = 770 Vdc; Ib2 = 60 µAdc; Eb1 at focus	t	500	----	hrs
Life-test end points: Linewidth “A” Modulation	----- 5226 5223		Ib2 = 100 µAdc; Ib2= 100 µAdc;	Width ΔEc1	----- -----	0.38 32	mm Vdc
Periodic-check tests							
Pressure (implosion)	1141						
Vibration	5111			Width		2	mm
Electrode current (grid No. 2)	5201		Ec1 = 0	Ic2	- 15	+ 15	µAdc
Cathode illumination	5216			---	----	----	----
Direct-interelectrode capacitance	1331		Cathode to all Control grid to all	Ck Cg1	---- ----	8 10	pF pF
Base material insulating quality	1216		----	-----	-----	-----	-------

NOTES:

1. When Ec2 is greater than 330 Vdc, Rg1 shall not exceed 0.5 MegΩ. When Ec2 is less than 330 Vdc, Rg1 shall not Exceed 1.5 MegΩ.

2. Cylinder shall be 4 inches (101.6 mm) long and 0.901 inch (22.89 mm) maximum inside diameter.

3. The screen characteristics shall be measured with a constant beam energy of 0.24 watt defocused to a spot approximately 0.25 cm in diameter.
   The test conditions shall be Eb2 = 10,000 Vdc.
   Ib2 = 24 µ Adc.
   The screen characteristics shall be cB5 – 450 cB minimum,
   G5:1 = 4 minimum

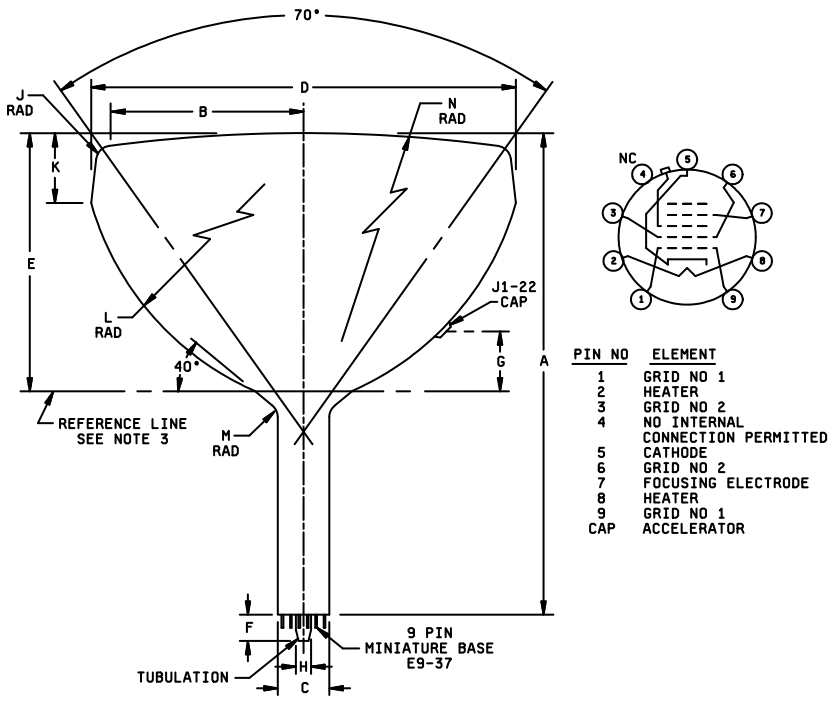
4. This test to be performed at the conclusion of the holding period.

5. The two sets of neck support referenced in method 5101 shall be spaced 1.125 inches (28.58 mm) and 3.25 inches (82.55 mm) below reference plane.

6. Tube type 7AQP33 has been added to this tube specification sheet.

7. An accept on zero defect sampling plan (see MIL-PRF-1, table III) shall be used for conformance inspection, part 1..

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parenthesis.
3. Reference line is determined by the point where leading edge of JEDEC-G-123 reference line gage will stop.

Ltr	Dimensions are in inches with metric equivalents (mm) in parentheses.	
	Minimum	Maximum
Conformance inspection, part 2		
A	7.88 (200.15)	8.25 (209.55)
B	3.00 (76.20)	----------
C	.81 (20.57)	.90 (22.86)
D	6.94 (176.28)	7.06 (179.32)
E	4.22 (107.19)	4.41 (112.01)
F	----------	.44 (11.18)
G	.84 (21.34)	1.09 (27.69)
H	----------	.25 (6.35)
Ltr	Dimensions are in inches with metric equivalents (mm) in parentheses	
J	.25 (6.35)	
K	1.12 (28.45)	
L	4.88 (123.95)	
M	.12 (3.05)	
N	24.00 (608.16)	

FIGURE 1. Outline drawing of electron tube type 7AQP7 and 7AQP33.

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Referenced documents. In addition to MIL-PRF-1, this document references the following:
MIL-STD-1311

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Navy – EC
DLA – CC

Preparing activity:
DLA- CC
Project number (5960-2008-070)

Review activities:
Navy – CG, MC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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